Symmtries in Wick Rotated QFT

Minjae V. Kim and Nathan Dressler

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These lecture notes for the Field Theory and Topology Reading Seminar at the University of Pennsylvania in Fall 2026 (https://maroon-scorch.github.io/ftt-2026/) is intended to both follow closely and expand on Lecture 3 of [F].

1 Axiom System for QFT

The goal of this lecture is to motivate the following Axiom system of a QFT by studying the progression

$$\mathbb{M}^n \rightsquigarrow \mathcal{D} \rightsquigarrow \mathbb{E}^n \rightsquigarrow X^n.$$

That is, the progression from a QFT with certain symmetries H_n, defined on affine Minkowski space $\mathbb{M}^n$, to a curved geometry X^n with differential H_n-structure.

Axiom 1 (Wick-rotated Field Theory)**.** A **field theory** is a symmetric monoidal functor

$$F : \mathrm{Bord}_{\langle n-1,n\rangle}(\mathcal{X}_n^{\nabla}) \to t\mathrm{Vect}_{\mathbb{C}} \tag{1}$$

Where $\mathcal{X}_n^{\nabla}$ is a geometric structure such as a Riemannian metric, conformal structure, section of a fiber bundle, connection on a principal bundle, topological structures, etc. These are commonly referred to as **background fields** in physics. For a bordism $X : Y_0 \to Y_1$:

- For a closed $(n-1)$-manifold Y_0, the functor assigns a topological vector space

  $$F(Y_0) \in \mathrm{tVect}_{\mathbb{C}},$$

  which is interpreted as the state space associated to the spatial manifold Y_0. In a more analytic formulation of QFT, one would typically impose additional structure to make this a Hilbert space.

- For an n-dimensional bordism $X : Y_0 \to Y_1$, the functor assigns a continuous linear map

  $$F(X) : F(Y_0) \longrightarrow F(Y_1),$$

  which can be interpreted as an evolution operator taking states on Y_0 to states on Y_1 [BMS].

- If two bordisms can be composed,

  $$Y_0 \xrightarrow{X_1} Y_1 \xrightarrow{X_2} Y_2,$$

  then functoriality gives

  $$F(X_2 \circ X_1) = F(X_2) \circ F(X_1)$$

  Thus, gluing bordisms corresponds to composing the associated evolution maps.

- For a disjoint union of $(n-1)$-manifolds, $Y_0 \sqcup Y_1$, the symmetric monoidal structure gives
$$F(Y_0 \sqcup Y_1) \cong F(Y_0) \otimes F(Y_1)$$
Thus disjoint systems correspond to tensor products of their state spaces.

- For a closed n-manifold X, viewed as a bordism $X : \varnothing \to \varnothing$, we have
$$F(X) : F(\varnothing) \longrightarrow F(\varnothing).$$
Since $F(\varnothing) \cong \mathbb{C}$ and every linear endomorphism of $\mathbb{C}$ is multiplication by a unique scalar, this is determined by a complex number $F(X) \in \mathbb{C}$, which is interpreted as the partiation function on X.

> *Remark* 1.1. The following points gives the geometric intuition for how correlation functions and local operators arise from the functorial description. Making this precise requires additional structure beyond the present axiom, such as a suitable theory of observables or factorization structure. In particular, the limit
> $$\varprojlim_{\epsilon \to 0} F(S_\epsilon(x))$$
> should, at this stage, be regarded as schematic notation rather than as a construction that follows forma=lly from the definition of F.

- To obtain correlation functions, let X be a closed n- manifold with points $x_1, \ldots, x_k \in X$ and remove small open balls of radius ϵ around these points forming
$$X_\epsilon = X \setminus \bigcup_{i=1}^{k} B_\epsilon(x_i)$$
Each boundary component is a sphere of dimension $n-1$ written $S_\epsilon(x_i) \cong S^{n-1}$. Taking all boundary componenets to be incoming, X_ϵ is a bordism
$$X_\epsilon : S_\epsilon(x_1) \sqcup \cdots \sqcup S_\epsilon(x_k) \longrightarrow \varnothing$$
The functor then gives a multilinear map
$$F(X_\epsilon) : F(S_\epsilon(x_1)) \otimes \cdots \otimes F(S_\epsilon(x_k)) \longrightarrow \mathbb{C}$$
In the inverse limit $\epsilon \to 0$, evaluating this map on local operators gives the Wick-rotated correlation function
$$\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_k(x_k) \rangle = \varprojlim_{\epsilon \to 0} F(X_\epsilon)\big(\mathcal{O}_1 \ldots, \mathcal{O}_k\big).$$

- The local operators (or observables) at point x_i are obtained by shrinking the surrounding sphere to a point:
$$\mathcal{O}_{x_i} \in \varprojlim_{\epsilon \to 0} F(S_\epsilon(x_i))$$
Thus the vector space $\lim_{\epsilon \to 0} F(S_\epsilon(x_i))$ is interpreted as the vector space of local operators at x_i.

- Hence the functor F packages the physical data of the theory into geometric and algebraic structures:
$$\begin{array}{ccc} \text{spatial manifold } Y & \longmapsto & \text{state space } F(Y), \\ \text{bordism } X : Y_0 \to Y_1 & \longmapsto & \text{evolution map } F(X), \\ \text{closed space } X & \longmapsto & \text{partition function } F(X), \\ \text{punctured spacetime } X_\epsilon & \longmapsto & \text{correlation functions}, \end{array}$$

Remark 1.2. The renormalization group in QFT can be realized by a family F_Ω of the functors in Axiom 1 for $\Omega \in \mathbb{R}^{>0}$ by composing F with the family of automorphisms of $\mathrm{Bord}_{\langle n-1,n\rangle}(\mathcal{X}_n^\nabla)$ which scale the Riemannian metric $g \mapsto \Omega^2 g$.

Remark 1.3. Although physical spacetime is typically noncompact, the bordism formalism is naturally formulated using compact manifolds. A compact bordism represents a finite spacetime process with specified boundary data. Noncompact spacetimes can then be treated by taking limits, imposing asymptotic boundary conditions, or considering an appropriate completion of the bordism category.

2 QM as a Quantum System (Reminder)

2.1 Reminder of QM

The axiomatic system of QM, in the first presented form, consists of following data:

Statement 2.1 (Mechanics)**.** Axiomatic system of a mechanical system has

1. States: The space of states is
$$\mathcal{S} = \left\{\sigma \in \mathcal{L}^1(\mathcal{H}) \,\middle|\, \sigma = \sigma^*,\ \sigma \geq 0,\ \mathrm{Tr}(\sigma) = 1\right\}.$$
The pure states are the rank-one orthogonal projections, and hence $\mathcal{S}_0 \cong \mathbb{P}\mathcal{H}$. In physics, σ is often called as the density matrix of a mixed state (although often $\mathcal{H}$ is not finite dimensional).

2. Operators: $\mathcal{O}$ are operators of $\mathcal{H}$. Observables are self-adjoint (possibly unbounded, densely defined) operators on $\mathcal{H}$:
$$\mathcal{O}_\mathbb{R} = \{A : A = A^*\}.$$
More generally, one considers the complex space of operators equipped with the involution
$$A \longmapsto A^*.$$

3. Hamiltonian: $H \in \mathcal{O}_\mathbb{R}^\infty$ where $\mathcal{O}^\infty$ is a dense subset of $\mathcal{O}$ with Lie algebra structure
$$[A_1, A_2]_\mathrm{P} = -\frac{i}{\hbar}(A_1A_2 - A_2A_1).$$
Deemed the Poisson Bracket.

4. Probability pairing: $p : \mathcal{O}_\mathbb{R} \times \mathcal{S} \to \mathrm{Prob}(\mathbb{R}),\ (A, \sigma) \mapsto \sigma_A$

5. Function on operators: $\mathrm{Borel}(\mathbb{R}) \times \mathcal{O}_\mathbb{R} \to \mathcal{O}_\mathbb{R}$ for $f \in \mathrm{Borel}(\mathbb{R})$ and $A \in \mathcal{O}_\mathbb{R}$ pairing to $f(A)$

6. Hamiltonian evolution: The self-adjoint Hamiltonian H generates the one-parameter unitary group
$$U_t = e^{-itH/\hbar}.$$
This induces the evolution $\sigma \longmapsto U_t \sigma U_t^*$ on states and $A \longmapsto U_t A U_t^*$ on observables.

Satisfying several axioms [F].

Remark 2.2. Especially, Hamiltonian H induces time evolution $t \mapsto U_t = \exp(-iHt/\hbar)$ in both $\mathcal{S}$ (Schrodinger picture) and $\mathcal{O}$ (Heisenberg picture). These satisfy certain compatibility conditions so that either picture can be chosen without changing the probability pairing, as a function of time.

For the discussions of QFT, it is useful to adopt an alternative axiomatic system, where we place the operator on top of the time $\mathbb{M}^1$ as a complex vector bundle.

Statement 2.3 (Alternative System)**.** An alternative Axiomatic system can be given as

2'. Operators with support: operators are given as a complex vector bundle with real structures $\mathcal{O} \to \mathbb{M}^1$ and a dense subbundle $\mathcal{O}^\infty$ .... Lie algebra structure. Call $\mathcal{O}_t$ as a fiber over time t.

4'. Correlation functions: Consider a finite sequence of time $t_1 < t_2 < \cdots < t_n$. A correlation function is a map $\langle\ \rangle : \mathcal{O}_{t_1} \times \mathcal{O}_{t_2} \times \cdots \times \mathcal{O}_{t_n} \times \mathcal{S} \to \mathbb{C}$. Its evaluation is denoted as $\langle A_{t_1} \dots A_{t_n} \rangle_\sigma \in \mathbb{C}$.

Remark 2.4. The probability paring is expected to be recoverable from the one-point correlation function via $\langle A \rangle_\sigma = \int \lambda \mathrm{d}\sigma_A(\lambda)$.

Remark 2.5. Correlation functions are not always well defined, e.g. multiplication of unbounded operators.

3 Symmetry of Relativistic QFT

3.1 Relativistic QFT

We begin from the structure of spacetime. Isometries of the spacetime is part of the data that we need to determine the symmetry of a QFT. Relativistic QFT takes place in Minkowski spacetime $\mathbb{M}^n$:

Definition 3.1 (Minkowski spacetime)**.** A **Minkowski spacetime** of $n = d+1$ dimension is an affine space $\mathbb{A}^n$ endowed with metric $\langle x, x \rangle = (x_0)^2 - (x^1)^2 - \cdots - (x^d)^2 = \eta_{\mu\nu} x^\mu x^\nu$

Consider a **relativistic QFT**, denoted as $\mathcal{Q}$, with a symmetry group $H_{1,n-1}$. We make an assumption that $H_{1,n-1}$ acts on the spacetime, **preserving the time orientation** via $\rho_n : H_{1,n-1} \longrightarrow \mathcal{I}^\uparrow_{1,n-1}$ and that the image of ρ_n **contains** the identity component $SO^\uparrow_{1,n-1}$. It has following properties/data:

1. **Fields** in $\mathcal{Q}$ are defined by a finite-dimensional $\mathbb{Z}_2$ graded Real reps

$$\sigma : H_{1,n-1} \longrightarrow \mathrm{Aut(R)} \tag{2}$$

Classical Fields are the functions $\mathbb{M}^n \longrightarrow R$. **Quantum Fields** are the R-valued Operator distributions $\Phi = \Phi^0 + \Phi^1$ on $\mathbb{M}^n$.

2. **N-point function** is a distribution, which takes a set of Schwartz functions $f_i : \mathbb{M}^n \longrightarrow R^*$ as an input and gives value in $\mathbb{C}$. For an example, a two-point function may be written as

$$\langle \Phi(f_1)\Phi(f_2) \rangle = \int_{(\mathbb{M}^n)^2} \mathrm{d}x_1 \mathrm{d}x_2\ f(x_1) f(x_2) \langle \Phi(x_1)\Phi(x_2) \rangle \tag{3}$$

where $\langle \Phi(x_1)\Phi(x_2)\rangle$ is the kernel of the $R_{\mathbb{C}}^{\otimes 2}$ valued distribution.

3. From the correlation functions, one can construct the $\mathbb{Z}/2$-graded **Hilbert space** $\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1$ (bosonic and fermionic states) which includes a vacuum state Ω.

4. The **field operators** $\Phi(f)$ act as unbounded operators on $\mathcal{H}$. In particular, 2-point functions compute the vacuum expectation value of the operators

$$\langle \Phi(f_1)\Phi(f_2)\rangle = \langle \Omega, \Phi(f_1)\Phi(f_2)\Omega\rangle_{\mathcal{H}} \tag{4}$$

5. An **unitary representation** $U\colon \mathcal{G}_{1,n-1} \to U(\mathcal{H})$ with $U(g)\Omega = \Omega$. In theories *without* supersymmetry, the image of U preserves the grading of $\mathcal{H}$. Both n-point functions and the vacuum state are **invariant** under the actions.

6. Positivity of energy: the spectral measure of $U|_{\mathbb{R}^{1,n-1}}$ is supported in the closed forward light cone dual to $\mathbb{R}_+^{1,n-1}$.

We additionally assumes the **Spin Statistics Theorem**. The action of $H_{1,n-1}$ preserves the $\mathbb{Z}_2$ grading of $\mathcal{H}$. In particular, $k_0 \in K \subset H_{1,n-1}$ acts as a grading operator of R.

3.2 Wick Rotation

A central idea of CRT comes from the observation that the correlation functions of sufficiently well behaving QFT can be analytically continued to the **complexified Minkowski spacetime**. The complexification involves **Wick rotation** of timelike vectors to the imaginary time.

Recall that a Minkowski spacetime is an affine space $\mathbb{M}^n$ endowed with a translation vector space $V = \mathbb{R}^{1,n-1}$, with a choice of future timelike cone $V_+ = \{\zeta : \langle \zeta, \zeta\rangle > 0\}$. We fix an orthogonal splitting $V = U \oplus U^\perp$.

Definition 3.2 (Complexified Minkowski Spacetime)**.** A complexified Minkowski spacetime $\mathbb{M}_{\mathbb{C}}^n$ is defined as an affine space $\mathbb{M}^n \times_V V_{\mathbb{C}}$. Its translation vector space is $V_{\mathbb{C}}$. As a set of points, it is $(\mathbb{M}^n \times V_{\mathbb{C}})/\sim$, where the orbits under $V \subset V_{\mathbb{C}}$ are quotiented out: $v \cdot (m, z) = (m + v, z - v) \sim (m, z)$.

It is helpful to consider $\mathbb{M}_{\mathbb{C}}^n$ as an affine space $V_{\mathbb{C}}$ over $V_{\mathbb{C}}$, where $V_{\mathbb{C}}$ is spanned by orthonormal vectors $e_0, e_1, \ldots, e_{n-1}$. Then, the Euclidean space $\mathbb{E}^n$ is an *Real* affine subspace as an orbit of $V_E = \langle e_0, e_1, \ldots, e_{n-1}\rangle_{\mathbb{R}}$ and the Minkowski space $\mathbb{M}^n$ is an *Real* affine subspace as an orbit of $V = \langle ie_0, e_1, \ldots, e_{n-1}\rangle_{\mathbb{R}}$.

What is the domain $\mathcal{D}$ to which we cal analytically continue the QFT? We start by observing the fact that the positive definite energy affords us to analytically continue to the **backward tube**.

$$\mathcal{T} \coloneqq V - iV_+ = \{(i(t - i\tau), \mathbf{x} - i\mathbf{y}) : (t, \mathbf{x}), (\tau, \mathbf{y}) \in \mathbb{M}^n, \text{where } |\mathbf{y}| < \tau\} \tag{5}$$

The correlation function in V is invariant under $SO_{1,n-1}^{\uparrow}$, and this invariance extends to $\mathcal{T}$, as it acts on V_+ too. That is, the correlation function is constant on the orbits of $SO_{1,n-1}^{\uparrow}$ acting on $\mathcal{T}$. We extend this along the orbits of the complexification $SO_n(\mathbb{C})$. The resulting domain is then

$$\mathcal{D} = SO_n(\mathbb{C})(\mathcal{T}) = O_n(\mathbb{C})(\mathcal{T}) \tag{6}$$

Remark 3.3. $\mathcal{D}$ includes both V and $V_E \setminus \{0\}$.

For the rest of this section, we call two point correlation function in $\mathbb{M}^n$ as

$$W(\xi) = \langle \Phi(x+\xi)\Phi(x) \rangle \tag{7}$$

and its complex extension as $W_{\mathbb{C}}$. Note that $W(\xi)$ for spacelike ξ can be acquired as a boundary value of $W_{\mathbb{C}}$, approaching from "$-i$ timelike" direction. That is,

$$W(\xi) = \lim_{\epsilon \to 0^+} W_{\mathbb{C}}(\xi - i\epsilon\eta), \qquad \xi \in V, \qquad \eta \in V_+ \ . \tag{8}$$

Remark 3.4. Note that in physics literature, W is generally written as a function of some preferred coordinates. Wick rotation is achieved by relabeling $t = -i\tau$ with $\tau > 0$. Then, the statement on the boundary value is that

$$W(t, \mathbf{x}) = \lim_{\epsilon \to 0^+} W_{\mathbb{C}}(t - i\epsilon, \mathbf{x}) \tag{9}$$

4 Symmetry Groups in QFT

With the notion of Wick Rotations in QFT, we may now study the subprogression

$$\mathbb{M}^n \rightsquigarrow \mathcal{D} \rightsquigarrow \mathbb{E}^n.$$

For a theory defined on Minkowski spacetime, we denote $\mathcal{G}_{1,n-1}$ to be the unbroken global symmetry group of the theory. We denote $\mathcal{I}_{1,n-1}$ to be the group of isometries of the spacetime, and K to be the symmetry group that leaves the spacetime invariant. We call K the internal symmetry group and assume K is compact following [F]. Heuristically, you may think:

$$\mathcal{G}_{1,n-1} \sim K \ \rtimes \mathcal{I}_{1,n-1}.$$

We will restrict to $\mathcal{I}^{\uparrow}_{1.n-1} \subset \mathcal{I}_{1.n-1}$, the subgroup of time orientation-preserving isometries. A common approach in quantum field theory, particularly in treatments based on the S-matrix, assumes that the Poincaré group is a subgroup of $\mathcal{G}_{1,n-1}$. The Coleman–Mandula theorem then implies, at the level of Lie algebras, a decomposition

$$\mathfrak{g}_{1,n-1} \cong \mathfrak{p}_{1,n-1} \oplus \mathfrak{k},$$

where $\mathfrak{p}_{1,n-1}$ is the Lie algebra of the Poincaré group and $\mathfrak{k}$ is the Lie algebra of K.

Instead, [F] puts forth the following hypothesis:

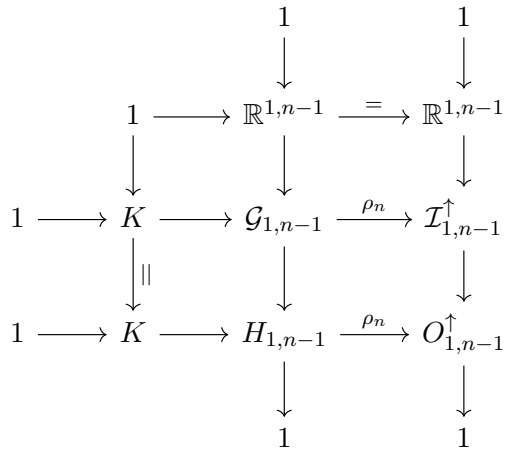
Statement 4.1. There exists a homomorphism $\mathcal{G}_{1,n-1} \to \mathcal{I}_{1,n-1}$ such that:

1. The image of the homomorphism contains the identity component $\mathcal{I}^0_{1,n-1}$
2. The kernel of the homomorphism is a compact Lie group K
3. The translation subgroup $\mathbb{R}^{1,n-1} \subset \mathcal{I}^{\uparrow}_{1,n-1}$ lifts to a normal subgroup of $\mathcal{G}_{1,n-1}$.

Still in this new approach, the kernel K is the group of internal symmetries of the theory; These are the symmetries that fix spacetime point-wise.

Definition 4.2. Define $H_{1,n-1} = \mathcal{G}_{1,n-1}/\mathbb{R}^{1,n-1}$, the reduced global symmetry group.

This produces the following diagram:



Note this has all been done in Minkowski spacetime $\mathbb{M}^n$. By the previous argument made in the Wick rotation section, we may complexify $H_{1,n-1}$ to act as symmetries on the complex correlation functions in the complexified domain $\mathcal{D}$. We define a different real form H_n acting on the Wick-rotated Euclidean theory $\mathbb{E}^n$ by choosing a new real slice of this complexified domain. This produces the following commutative diagram of Lie groups:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & K & \longrightarrow & H_{1,n-1} & \xrightarrow{\rho_n} & O^{\uparrow}_{1,n-1} & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & K(\mathbb{C}) & \longrightarrow & H_n(\mathbb{C}) & \xrightarrow{\rho_n} & O_n(\mathbb{C}) & \longrightarrow & 1 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
1 & \longrightarrow & K & \longrightarrow & H_n & \xrightarrow{\rho_n} & O_n & \longrightarrow & 1
\end{array}$$

Remark 4.3. The kernel of $\rho_n : H_n \longrightarrow O_n$ is the same compact real form $K \subset K(\mathbb{C})$ as the kernel of

$$\rho_n : H_{1,n-1} \longrightarrow O^{\uparrow}_{1,n-1}.$$

This is simply because the internal symmetries are defined to act trivially on spacetime, and are therefore unaffected by the change from Minkowski to Euclidean spacetime. Thus the Euclidean real form of the kernel is taken to be the same compact real form $K \subset K(\mathbb{C})$.

Remark 4.4. This implies that the problem of classifying the **symmetry** of a Relativistic QFT is equivalent to the problem of classifying the **symmetry type** (H_n, ρ_n).

Remark 4.5. Since the identity component of O_n is SO_n, $SO_n \subseteq \mathrm{im}(\rho_n) \subseteq O_n$, and SO_n has index two in O_n, there are only two possibilities: $\mathrm{im}(\rho_n) = SO_n$ or $\mathrm{im}(\rho_n) = O_n$. Thus the Euclidean theory either has only orientation-preserving spacetime symmetries, or it also has reflections. After anti-Wick rotation, these correspond respectively to the absence or presence of the corresponding time-reversing symmetries.

Let $SH_n = \rho^{-1}(SO_n)$ and $\widetilde{SH_n}$ be the double cover of SH_n. Note this double cover of SH_n can be constructed using Spin_n, the double cover of SO_n. That is:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & K & \longrightarrow & \widetilde{SH_n} & \longrightarrow & \mathrm{Spin}_n & \longrightarrow & 1 \\
 & & \| & & \downarrow{\scriptstyle 2:1} & & \downarrow{\scriptstyle 2:1} & & \\
1 & \longrightarrow & K & \longrightarrow & SH_n & \longrightarrow & SO_n & \longrightarrow & 1 \\
 & & \| & & \downarrow{\scriptstyle 1:2} & & \downarrow{\scriptstyle 1:2} & & \\
1 & \longrightarrow & K & \longrightarrow & H_n & \xrightarrow{\rho_n} & O_n & \longrightarrow & 1
\end{array}$$

If $\rho_n : H_n \to O_n$ is surjective, then we define $\widetilde{H_n}$ in the following way:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & K & \longrightarrow & \widetilde{H_n} & \longrightarrow & \mathrm{Spin}_n & \longrightarrow & 1 \\
 & & \| & & \downarrow{\scriptstyle 2:1} & & \downarrow{\scriptstyle 2:1} & & \\
1 & \longrightarrow & K & \longrightarrow & H_n & \xrightarrow{\rho_n} & O_n & \longrightarrow & 1
\end{array}$$

Let $\mathfrak{h}_\mathfrak{n}, \mathfrak{k}, \mathfrak{o}_\mathfrak{n}$ be the Lie algebras of H_n, K, O_n respectfully.

Theorem 4.6 (Theorem 3.16)**.** *Assume $n \geq 3$.*

1. *There is a splitting*
$$\mathfrak{h}_n = \mathfrak{o}'_n \oplus \mathfrak{k},$$
and ρ_n induces an isomorphism of Lie algebras
$$\mathfrak{o}'_n \xrightarrow{\cong} \mathfrak{o}_n.$$

2. *If $n \geq 3$, there is an isomorphism*
$$\widetilde{SH}_n \cong \mathrm{Spin}_n \times K.$$
Hence there exists a central element $k_0 \in K$ with $k_0^2 = 1$ and an isomorphism
$$SH_n \cong \mathrm{Spin}_n \times K / \langle (-1, k_0) \rangle,$$
where $\langle (-1, k_0) \rangle$ is the cyclic group generated by $(-1, k_0)$.

3. *There is a canonical homomorphism*
$$\mathrm{Spin}_n \longrightarrow H_n$$
under which the image of the central element $-1 \in \mathrm{Spin}_n$ is $k_0 \in K$.

4. *If $n \geq 3$ and $\rho_n : H_n \longrightarrow O_n$ is surjective, then there exists a group extension*
$$1 \longrightarrow K \longrightarrow J \longrightarrow \{\pm 1\} \longrightarrow 1$$
and a pullback diagram of group extensions
$$\begin{array}{ccccccccc}
1 & \longrightarrow & K & \longrightarrow & \widetilde{H}_n & \xrightarrow{\rho_n} & \mathrm{Pin}_n^+ & \longrightarrow & 1 \\
 & & \| & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & K & \longrightarrow & J & \longrightarrow & \{\pm 1\} & \longrightarrow & 1.
\end{array}$$
There is an isomorphism
$$H_n \cong \widetilde{H}_n / \langle (-1, k_0) \rangle.$$

Note that (1.) states [F]'s version of the Coleman–Mandula theorem. Essentially, it says that H_n is locally modeled on the product $K \times O_n$: infinitesimally, the spacetime and internal symmetries separate.

Part (2.) is the corresponding splitting at the level of Lie groups rather than Lie algebras: $\widetilde{SH}_n \cong \mathrm{Spin} \times \mathrm{K}$ after passing to the spin double cover. However, SH_n itself is not necessarily a direct product since the central element $-1 \in \mathrm{Spin_n}$ is identified with a central element $k_0 \in K$. In other words,

$$SH_n \cong (\mathrm{Spin}_n \times K)/\langle(-1, k_0)\rangle.$$

Part (3.) the homomorphism $\mathrm{Spin}_n \to H_n$ of Theorem 4.4 should be viewed as describing the Lorentz (spin) part of the spacetime symmetry after passing to the quotient by translations. Under anti-Wick rotation, the Euclidean group Spin_n analytically continues to the Lorentz spin group $\mathrm{Spin}_{1,n-1}$, while the lifted translation subgroup $\mathbb{R}^{1,n-1}$ is unchanged. Thus, restoring the translations that were quotiented out in defining H_n, the map of Theorem 4.4 gives heuristically

$$\mathbb{R}^{1,n-1} \rtimes \mathrm{Spin}_{1,n-1} \longrightarrow \mathcal{G}_{1,n-1},$$

i.e. the usual action of the (spin cover of the) Poincaré group on the full symmetry group of the Minkowski theory. In particular, we can produce the usual construction in QFT which assumes that the Poincaré group is a subgroup of $\mathcal{G}_{1,n-1}$ mentioned earlier.

Finally, Part (4.) describes the additional structure that appears when orientation-reversing symmetries are present. In this case the orientation-reversing part is controlled by a group extension

$$1 \longrightarrow K \longrightarrow J \longrightarrow \{\pm 1\} \longrightarrow 1,$$

which is independent of n. Thus the connected symmetry group splits after passing to the spin double cover, while the extension J records the additional information carried by orientation-reversing symmetries.

Remark 4.7. For part 2, in the event $n = 2$, we only get

$$\widetilde{SH}_2 \cong \mathrm{Spin}_2 \rtimes K.$$

The full proof of Theorem 4.3 is contained in [FH]. Though we give brief sketches of parts 1 and 2:

Proof of Theorem 3.16(1). We give the main Lie-algebraic idea. Since H_n is a compact Lie group, its Lie algebra $\mathfrak{h}_n$ is reductive, so

$$\mathfrak{h}_n = [\mathfrak{h}_n, \mathfrak{h}_n] \oplus \mathfrak{z}(\mathfrak{h}_n),$$

where the derived algebra $[\mathfrak{h}_n, \mathfrak{h}_n]$ is semisimple and $\mathfrak{z}(\mathfrak{h}_n)$ is the center. In particular, the Killing form of $[\mathfrak{h}_n, \mathfrak{h}_n]$ is nondegenerate. Since $n \geq 3$, the Lie algebra $\mathfrak{o}_n$ is semisimple, and in particular has zero center. The differential $\rho_n : \mathfrak{h}_n \to \mathfrak{o}_n$ is surjective with kernel $\mathfrak{k}$. Because $\mathfrak{o}_n$ is semisimple, $[\mathfrak{o}_n, \mathfrak{o}_n] = \mathfrak{o}_n$, and hence

$$\rho_n\big([\mathfrak{h}_n, \mathfrak{h}_n]\big) = [\mathfrak{o}_n, \mathfrak{o}_n] = \mathfrak{o}_n.$$

Set $\mathfrak{k}_{ss} := \mathfrak{k} \cap [\mathfrak{h}_n, \mathfrak{h}_n]$, which is an ideal of $[\mathfrak{h}_n, \mathfrak{h}_n]$ since $\mathfrak{k} = \ker \rho_n$ is an ideal of $\mathfrak{h}_n$. Let $\mathfrak{o}'_n$ be the orthogonal complement of $\mathfrak{k}_{ss}$ in $[\mathfrak{h}_n, \mathfrak{h}_n]$ with respect to the Killing form. Since the Killing form is nondegenerate and $\mathfrak{k}_{ss}$ is an ideal of a semisimple Lie algebra, $\mathfrak{o}'_n$ is also an ideal and

$$[\mathfrak{h}_n, \mathfrak{h}_n] = \mathfrak{o}'_n \oplus \mathfrak{k}_{ss}.$$

The restriction $\rho_n|_{\mathfrak{o}'_n} : \mathfrak{o}'_n \to \mathfrak{o}_n$ is injective, because its kernel lies in $\mathfrak{o}'_n \cap \mathfrak{k}_{ss} = 0$. It is surjective, because

$$\mathfrak{o}_n = \rho_n\big([\mathfrak{h}_n, \mathfrak{h}_n]\big) = \rho_n(\mathfrak{o}'_n) + \rho_n(\mathfrak{k}_{ss}) = \rho_n(\mathfrak{o}'_n).$$

Hence $\rho_n|_{\mathfrak{o}'_n}$ is an isomorphism of Lie algebras. Finally, ρ_n maps $\mathfrak{z}(\mathfrak{h}_n)$ into $\mathfrak{z}(\mathfrak{o}_n) = 0$ by surjectivity, so $\mathfrak{z}(\mathfrak{h}_n) \subseteq \mathfrak{k}$. Consequently $\mathfrak{k} = \mathfrak{k}_{ss} \oplus \mathfrak{z}(\mathfrak{h}_n)$: given $x \in \mathfrak{k}$, write $x = a + z$ with $a \in [\mathfrak{h}_n, \mathfrak{h}_n]$ and $z \in \mathfrak{z}(\mathfrak{h}_n) \subseteq \mathfrak{k}$; then $a = x - z \in \mathfrak{k} \cap [\mathfrak{h}_n, \mathfrak{h}_n] = \mathfrak{k}_{ss}$. Therefore

$$\mathfrak{h}_n = \mathfrak{o}'_n \oplus \mathfrak{k}_{ss} \oplus \mathfrak{z}(\mathfrak{h}_n) = \mathfrak{o}'_n \oplus \mathfrak{k}.$$

Thus, infinitesimally, the spacetime and internal symmetries separate. □

Proof of Theorem 3.16(2). For $n \geq 3$, Part (1) gives a splitting of Lie algebras

$$\mathfrak{sh}_n \cong \mathfrak{o}'_n \oplus \mathfrak{k}, \qquad \mathfrak{o}'_n \cong \mathfrak{so}_n.$$

Passing to the simply connected cover removes the possible global obstruction to integrating this splitting. Since Spin_n is the simply connected Lie group with Lie algebra $\mathfrak{so}_n$, the $\mathfrak{o}'_n$-factor integrates to Spin_n, and, by the construction of $\widetilde{SH}_n$, the $\mathfrak{k}$-factor integrates to K. Hence

$$\widetilde{SH}_n \cong \mathrm{Spin}_n \times K.$$

Let $q : \widetilde{SH}_n \longrightarrow SH_n$ be the covering homomorphism. The kernel of this covering is the central subgroup of order 2, so under the above identification we may write $\ker(q) = \{(1,1), (-1,k_0)\}$ for some $k_0 \in K$. Since $(-1,k_0)$ has order 2,

$$(1,1) = (-1,k_0)^2 = (1,k_0^2),$$

and hence $k_0^2 = 1$. Moreover, since $\ker(q)$ is central, $(-1,k_0)$ is central in $\mathrm{Spin}_n \times K$, and therefore k_0 is central in K. Finally, the first isomorphism theorem gives

$$SH_n \cong \frac{\widetilde{SH}_n}{\ker(q)} \cong \frac{\mathrm{Spin}_n \times K}{\langle(-1,k_0)\rangle}.$$

□

Proposition 4.8 (Proposition 2.16)**.** *Assume $n \geq 3$. If the internal symmetry group K is trivial, then*

$$H_n \cong \mathrm{SO}_n \qquad \text{or} \qquad H_n \cong \mathrm{O}_n.$$

If $K \cong \{\pm 1\}$ is cyclic of order two, then there are six possibilities for H_n up to isomorphism:

$$\mathrm{SO}_n \times \{\pm 1\}, \qquad \mathrm{Spin}_n, \qquad \mathrm{O}_n \times \{\pm 1\}, \qquad E_n, \qquad \mathrm{Pin}_n^+, \qquad \mathrm{Pin}_n^-.$$

Proof. If $K = \{1\}$, then since $\mathrm{im}(\rho_n)$ is either SO_n or O_n, we have the first statement.

Now suppose $K \cong \{\pm 1\}$. Then H_n is a central extension

$$1 \longrightarrow \{\pm 1\} \longrightarrow H_n \xrightarrow{\rho_n} G \longrightarrow 1, \qquad G = \mathrm{SO}_n \text{ or } \mathrm{O}_n.$$

Such extensions are classified by

$$H^2(B\mathrm{SO}_n; \mathbb{Z}/2) \cong \mathbb{Z}/2$$

in the SO_n case, and by

$$H^2(B\mathrm{O}_n; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2$$

in the O_n case. Working out the corresponding extensions gives the six groups

$$\mathrm{SO}_n \times \{\pm 1\}, \quad \mathrm{Spin}_n, \quad \mathrm{O}_n \times \{\pm 1\}, \quad E_n, \quad \mathrm{Pin}_n^+, \quad \mathrm{Pin}_n^-.$$

In the first, third, and fourth cases, the nonidentity element of K is not the image of the central element $-1 \in \mathrm{Spin}_n$. The remaining cases give the five basic symmetry types below. □

States / symmetry	H_n	K	k_0
Bosons only	SO_n	$\{1\}$	$\mathrm{k}_0 = 1$
Fermions allowed	Spin_n	$\{\pm 1\}$	$\mathrm{k}_0 = -1$
Bosons, time-reversal (T)	O_n	$\{1\}$	$\mathrm{k}_0 = 1$
Fermions, $\mathrm{T}^2 = (-1)^F$	Pin_n^+	$\{\pm 1\}$	$\mathrm{k}_0 = -1$
Fermions, $\mathrm{T}^2 = \mathrm{id}$	Pin_n^-	$\{\pm 1\}$	$\mathrm{k}_0 = -1$

Table 1: The five basic symmetry types.

Below, we introduce an idea of a stabilization of H_n, the main structure result needed about symmetry groups in QFT.

Theorem 4.9 (Theorem 3.24)**.** *Assume $n \geq 3$. There exist compact Lie groups H_m, $m > n$, and homomorphisms i_n, ρ_n which fit into the commutative diagram*

$$\begin{array}{ccccccc} H_n & \xhookrightarrow{i_n} & H_{n+1} & \xhookrightarrow{i_{n+1}} & H_{n+2} & \hookrightarrow & \cdots \\ \downarrow{\scriptstyle \rho_n} & & \downarrow{\scriptstyle \rho_{n+1}} & & \downarrow{\scriptstyle \rho_{n+2}} & & \\ O_n & \hookrightarrow & O_{n+1} & \hookrightarrow & O_{n+2} & \hookrightarrow & \cdots \end{array} \tag{3.25}$$

in which squares are pullbacks. Furthermore, any two choices of stabilization the above are isomorphic.

Proof sketch. Theorem 3.16 shows that, for $n \geq 3$, the symmetry group H_n is determined by dimension-independent internal data. If $\mathrm{im}(\rho_n) = SO_n$, then

$$H_n \cong \frac{\mathrm{Spin}_n \times K}{\langle (-1, k_0) \rangle}.$$

We stabilize by using the standard inclusion $\mathrm{Spin}_n \hookrightarrow \mathrm{Spin}_{n+1}$ and the identity on K. This gives $H_n \hookrightarrow H_{n+1}$. If $\mathrm{im}(\rho_n) = O_n$, Theorem 4.3 instead gives a dimension-independent extension

$$1 \to K \to J \to \{\pm 1\} \to 1$$

and realizes $\widetilde{H}_n$ as the pullback of J along $\mathrm{Pin}_n^+ \to \{\pm 1\}$. We stabilize by using $\mathrm{Pin}_n^+ \hookrightarrow \mathrm{Pin}_{n+1}^+$ while keeping J fixed. Quotienting by $\langle (-1, k_0) \rangle$ gives $H_n \hookrightarrow H_{n+1}$. In both cases the construction is compatible with the standard inclusion $O_n \hookrightarrow O_{n+1}$ and the resulting squares are pullbacks. Since the data K, k_0, and, J are independent of n, any two such stabilizations are isomorphic. Thus

$$H_n \hookrightarrow H_{n+1} \hookrightarrow H_{n+2} \hookrightarrow \cdots$$

gives the desired stabilization. □

Note that $K = \ker \rho_n$ is independent of n. Thus this theorem allows us to extend the notion of symmetry groups in QFT to all dimensions. This creates a notion of "Spin Theories", "Pin$^+$ theories, "SO Theoreis" etc. independent of dimension by, for example, setting

$$H = \operatorname*{colim}_{n\to\infty} H_n$$

Definition 4.10 (Definition 3.27)**.** The homomorphism

$$\rho : H \longrightarrow O$$

is called the symmetry type and is denoted by (H, ρ).

The stabilization therefore allows us to regard the symmetry data as a single dimension-independent object (H, ρ) where

$$H = \operatorname*{colim}_{n\to\infty} H_n, \qquad \rho : H \longrightarrow O = \operatorname*{colim}_{n\to\infty} O_n.$$

The finite-dimensional symmetry types (H_n, ρ_n) are recovered by pullback along $O_n \hookrightarrow O$. Hence, we also use the term "symmetry type" for the pair (H_n, ρ_n) in our n-dimensional QFT.

Remark 4.11. The symmetry type (H, ρ) encodes the full symmetry group H together with its action on the spacetime symmetry group O. The finite-dimensional pairs (H_n, ρ_n) obtained by the pullback thus contain enough information to reconstruct the full symmetry type.

We can now return to the original goal of describing a quantum field theory geometrically. At the beginning of the lecture, a field theory was packaged as a symmetric monoidal functor

$$F : \mathrm{Bord}_{\langle n-1,n\rangle}(\mathcal{X}_n^{\nabla}) \longrightarrow \mathrm{tVect}_{\mathbb{C}}.$$

The role of the symmetry type is to constrain and organize the functorial data of the field theory. The bordism category tells us what geometric processes are possible, the functor F tells us what physical data those processes produce, and the symmetry type (H, ρ) tells us which spacetime and internal transformations preserve the theory. In this sense, the passage from Minkowski space through Wick rotation to a stable Euclidean symmetry type explains the symmetry data that the functorial description of a quantum field theory is required to respect. The stabilization allows us to regard these symmetry data as dimension-independent.

4.1 CRT

Proofs are provided in Appendix A. of [FH].

Statement 4.12 (Complex double covers of $O_n(\mathbb{C})$)**.** (Prop. A.11) $\mathrm{Pin}_n^{\pm}(\mathbb{C})$ are the unique complex Lie groups with identity component $\mathrm{Spin}_n(\mathbb{C})$ which double cover $O_n(\mathbb{C})$ and contain $\mathrm{Pin}_n^{\pm}$ as maximal compact subgroups. Any complex Lie group double covering $O_n(\mathbb{C})$ with identity component isomorphic to $\mathrm{Spin}_n(\mathbb{C})$ is isomorphic to $\mathrm{Pin}_n^{+}(\mathbb{C})$ or $\mathrm{Pin}_n^{-}(\mathbb{C})$.

Definition 4.13 (α- and β-double covers)**.** Let $\mu_m \subset \mathbb{C}^{\times}$ be the mth roots of unity, and

let $\pi_2 \colon \mathrm{Spin}_n(\mathbb{C}) \to SO_n(\mathbb{C})$ and $\pi_4 \colon \mathrm{Spin}_n(\mathbb{C}) \times_{\mu_2} \mu_4 \to SO_n(\mathbb{C})$ be the projections;

$$\begin{array}{ccc} \mathrm{Spin}_n(\mathbb{C}) & \hookrightarrow & \mathrm{Spin}_n(\mathbb{C}) \times_{\mu_2} \mu_4 \\ & {\scriptstyle \pi_2} \searrow \quad \swarrow {\scriptstyle \pi_4} & \\ & \mathrm{SO}_n(\mathbb{C}) & \end{array} \tag{10}$$

let

$$\begin{aligned} \widetilde{SO}^{\alpha}_{1,n-1} &:= \pi_2^{-1}(SO_{1,n-1}), \\ \widetilde{SO}^{\beta}_{1,n-1} &:= \mathrm{Spin}_{1,n-1} \ \cup \ \Big(\pi_4^{-1}(SO^{\downarrow}_{1,n-1}) \setminus \pi_2^{-1}(SO^{\downarrow}_{1,n-1})\Big). \end{aligned}$$

Similarly for $\mathrm{Pin}^{\pm}_n(\mathbb{C})$ over $O_n(\mathbb{C})$;

$$\begin{array}{ccc} \mathrm{Pin}^{\pm}{}_n(\mathbb{C}) & \hookrightarrow & \mathrm{Pin}^{\pm}{}_n(\mathbb{C}) \times_{\mu_2} \mu_4 \\ & {\scriptstyle \pi_{2\pm}} \searrow \quad \swarrow {\scriptstyle \pi_{4\pm}} & \\ & \mathrm{O}_n(\mathbb{C}) & \end{array} \tag{11}$$

let

$$\begin{aligned} \widetilde{O}^{\alpha}_{n-1,1} &:= \pi_{2+}^{-1}(O_{1,n-1}), \\ \widetilde{O}^{\beta}_{n-1,1} &:= \pi_{2+}^{-1}\big(O^{\uparrow}_{1,n-1}\big) \ \cup \ \Big(\pi_{4+}^{-1}\big(O^{\downarrow}_{1,n-1}\big) \setminus \pi_{2+}^{-1}\big(O^{\downarrow}_{1,n-1}\big)\Big), \end{aligned}$$

and

$$\begin{aligned} \widetilde{O}^{\alpha}_{1,n-1} &:= \pi_{2-}^{-1}(O_{1,n-1}), \\ \widetilde{O}^{\beta}_{1,n-1} &:= \pi_{2-}^{-1}\big(O^{\uparrow}_{1,n-1}\big) \ \cup \ \Big(\pi_{4-}^{-1}\big(O^{\downarrow}_{1,n-1}\big) \setminus \pi_{2-}^{-1}\big(O^{\downarrow}_{1,n-1}\big)\Big), \end{aligned}$$

Remark 4.14 (Difference of α- and β- double cover). Distinction of α and β comes from the order of time-reversal elements, such as $g = \mathrm{diag}(-1, -1, 1, \ldots, 1)$. In α double cover, such elements lift to an order 4 element. However, in the β double cover, such elements are paired with the fourth roots of unity, and therefore are of order 2.

Statement 4.15 (Classification of Lorentz double covers, Prop. A.15)**.** Lorentz double covers can thus be classified as follows:

1. Every double cover of $SO_{1,n-1}$ whose identity component is isomorphic to $\mathrm{Spin}_{1,n-1}$ is isomorphic to $\widetilde{SO}^{\alpha}_{1,n-1}$ or $\widetilde{SO}^{\beta}_{1,n-1}$.
2. $\widetilde{SO}^{\beta}_{1,n-1}$ is a subgroup of the even subalgebras of both $\mathrm{Cliff}_{n-1,1}$ and $\mathrm{Cliff}_{1,n-1}$.
3. $\widetilde{O}^{\beta}_{n-1,1}$ and $\widetilde{O}^{\beta}_{1,n-1}$ are subgroups of $\mathrm{Cliff}_{n-1,1}$ and $\mathrm{Cliff}_{1,n-1}$ respectively, and

$$\mathrm{Pin}_{n-1,1} \cong \widetilde{O}^{\beta}_{n-1,1}, \qquad \mathrm{Pin}_{1,n-1} \cong \widetilde{O}^{\beta}_{1,n-1}.$$

The α-double covers are subgroups of **complex (s)pin groups**; the β-double covers are subgroups of **Lorentz signature Clifford algebras**.

Definition 4.16 (α- and β-extensions of general symmetry types)**.** Let $(H_{1,n-1}, \rho_n)$ be a Lorentz signature symmetry type, $\rho_n \colon H_{1,n-1} \to O^{\uparrow}_{1,n-1}$, with internal group K and central element $k_0 \in K$ as in Todo: which theorem in our note?. Set

$$SH^{\alpha/\beta}_{1,n-1} := \widetilde{SO}^{\alpha/\beta}_{1,n-1} \times K \big/ \langle (-1, k_0) \rangle.$$

If $\operatorname{im} \rho_n = SO^{\uparrow}_{1,n-1}$, set $H^{\alpha/\beta}_{1,n-1} := SH^{\alpha/\beta}_{1,n-1}$. If ρ_n is surjective, define $\widetilde{H}^{\alpha/\beta}_{1,n-1}$ by the pullback diagram:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & K & \longrightarrow & \widetilde{H}^{\alpha/\beta}_{1,n-1} & \longrightarrow & \widetilde{O}^{\alpha/\beta}_{n-1,1} & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle ||} & & \downarrow & & \downarrow{\scriptstyle \det} & & \\
1 & \longrightarrow & K & \longrightarrow & J & \longrightarrow & \{\pm 1\} & \longrightarrow & 1 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 1 & & 1 & &
\end{array} \tag{12}$$

, and define

$$H^{\alpha/\beta}_{1,n-1} := \widetilde{H}^{\alpha/\beta}_{1,n-1} \big/ \langle (-1, k_0) \rangle.$$

One has $\mathrm{Spin}^{\alpha/\beta}_{1,n-1} \cong \widetilde{SO}^{\alpha/\beta}_{1,n-1}$, and $H^{\alpha}_{1,n-1}$ is a real subgroup of $H_n(\mathbb{C})$, namely the inverse image of $O_{1,n-1}$ under $\rho_n \colon H_n(\mathbb{C}) \to O_n(\mathbb{C})$.

Statement 4.17 (Extensions of real representations, Prop. A.20)**.** Let $R = R^0 \oplus R^1$ be a $\mathbb{Z}/2\mathbb{Z}$-graded real representation of $H_{1,n-1}$ on which $k_0 \in K$ acts as the grading operator, and let $R_{\mathbb{C}} = R \otimes_{\mathbb{R}} \mathbb{C}$, which carries an action of $H_n(\mathbb{C})$ and hence of $H^{\alpha}_{1,n-1}$.

1. If $h \in H^{\alpha}_{1,n-1} \setminus H_{1,n-1}$, then $h(R^0) = R^0$ and $h(R^1) = \sqrt{-1}\, R^1$.
2. The action of $H_{1,n-1}$ on R extends canonically to $H^{\beta}_{1,n-1}$.

All groups here are ungraded, so act by even transformations; the conclusion is that the β-extension acts on *real* representations of $H_{1,n-1}$.

Theorem 4.18 (CRT Theorem, [FH, Thm. A.23])**.** *Let $\mathcal{Q}$ denote a relativistic quantum field theory with symmetry group $H_{1,n-1}$. Then the symmetry extends to $H^{\beta}_{1,n-1}$; elements of $H^{\beta}_{1,n-1} \setminus H_{1,n-1}$ act antilinearly.*

Todo: State CRT –¿ although we assumed only the time order preserving symmetry actions, we end up with T element

Todo: CRT Proof...

5 Wick-Rotated Field Theory on Compact Manifolds

The last stage in our progression for this lecture is the transition from

$$\mathbb{E}^n \rightsquigarrow X^n,$$

that is, from Euclidean space to general curved manifolds. There is a standard differential-geometric construction which associates to a homomorphism

$$\rho_n : H_n \longrightarrow O_n$$

the notion of a differential H_n-structure on an n-manifold. We review this construction in Appendix B. For the purposes of this lecture, we apply this construction to the Euclidean field theory of the previous section. This allows us to replace Euclidean space by arbitrary curved manifolds equipped with the appropriate geometric structure, and leads naturally to the axiomatic formulation of field theory.

For now, we apply that same logic to our Euclidean field theory in section 4 to arrive at Axiom 1.

For a theory with symmetry type (H_n, ρ_n), the final ingredient is to replace a fixed background spacetime by arbitrary manifolds equipped with a differential H_n-structure. The appropriate domain for the Wick-rotated theory is therefore the bordism category

$$\mathrm{Bord}_{\langle n-1,n\rangle}(H_n^\nabla).$$

An object of $\mathrm{Bord}_{\langle n-1,n\rangle}(H_n^\nabla)$ is a closed $(n-1)$-manifold Y_0 together with the geometric structure determined by (H_n, ρ_n). A morphism

$$Y_0 \longrightarrow Y_1$$

is represented by an n-dimensional manifold X whose boundary is identified with

$$\partial X \cong \overline{Y_0} \sqcup Y_1,$$

together with a differential H_n-structure on X which restricts to the given structures on the boundary. Composition of morphisms is given by gluing bordisms along their common boundary.

As discussed earlier, the functor F encodes the basic physical data of the Wick-rotated theory. The vector spaces $F(Y)$ encode the spaces of states associated to spatial manifolds, while the linear maps associated to bordisms encode the corresponding amplitudes, evolution of state spaces, and correlation functions. Observables and their correlation functions can be extrapolated via the functorial structure with some additional structure not reviewed in this lecture.

This completes the progression:

$$\mathbb{M}^n \rightsquigarrow \mathcal{D}^n \rightsquigarrow \mathbb{E}^n \rightsquigarrow X^n(H_n^\nabla),$$

where $\mathbb{M}^n$ denotes Minkowski spacetime, $\mathcal{D}^n$ denotes the complexified setting, $\mathbb{E}^n$ denotes Euclidean space, and $X^n(H_n^\nabla)$ denotes a general curved n-manifold equipped with the appropriate differential H_n-structure.

Ideally, one would like a reconstruction theorem which reverses the progression done in this lecture showing that sufficiently general geometric and categorical data determine the original physical theory. Such a theorem would provide a precise mathematical explanation for how the usual formulation of quantum field theory on Minkowski spacetime can be recovered from the more abstract Axiom System. At present, however, this should be regarded as a structural goal rather than a general theorem.

A Real Pin group and Spin group

We can consider a group Hom $O_{1,n-1} \to \mathbb{Z}_2 \times \mathbb{Z}_2$ where the first part extracts the sign of $0,0$ component of the matrix and the second part extracts the sign of the determinant of the lower right $(n-1)\times(n-1)$ block (Isometry guarantees that neither is zero). From this, it is not too hard to realize that

$$O_{1,n-1} = SO^{\uparrow}_{1,n-1} \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

where $SO^{\uparrow}_{1,n-1}$ is the identity component, the first copy of $\mathbb{Z}_2$ is generated by $T \coloneqq \mathrm{diag}(-1,1,\dots 1)$ and the second copy is generated by $R \coloneqq \mathrm{diag}(1,-1,1,\dots 1)$.

Definition A.1 (Clifford Algebra)**.** Clifford algebra of a vector space V over field $\mathbb{K}$ (characteristic $\neq 2$), induced by a quadratic form q is

$$\mathrm{Cl}(V,q) = T(V)/(v\otimes v - q(v)) \tag{13}$$

where $T(V)$ is the tensor algebra of V.

For a Real Clifford algebra, we denote the signature of q as $\mathrm{Cl}_{a,b}$. For an example, $\mathrm{Cl}_{1,n-1}$ is defined from

$$e_0^2 = -1,\ e_1^2 = e_2^2 = \cdots = e_{n-1}^2 = 1$$

Definition A.2 (Clifford Group)**.** Define α to be the automorphism that is induced by -1 acting on the vector space V. That is, α acts by -1 on Cl^1 and 1 on Cl^0. The grading is decided by the degree (mod 2) of the spanning sets. The Clifford group Γ_V is the group of homogeneous invertible elements x of the Clifford algebra such that $\alpha(x)Vx^{-1} \subset V$. Also this defines the group action of Γ_V on V.

Remark A.3. An element $v \in V$ acts on V as a reflection through v.

Remark A.4. An element of Γ_V is a monomial of form $v_1 v_2 \dots v_n$. It acts trivially on V iff it is in $\mathbb{K}^\times$.

From a construction akin to Euler angle, it is evident that a O_V is generated by some rotations in 2d planes and reflections. Moreover, any rotations in 2d planes are generated by reflections. Therefore, O_V is generated by reflections. [1] Therefore, we have a central group extension:

$$1 \longrightarrow \mathbb{K}^\times \longrightarrow \Gamma_V \longrightarrow O_V \longrightarrow 1 \tag{14}$$

Definition A.5 (Spinor Norm)**.** Let $-^t$ be an anti-automorphism extended from the identity map of V, so that $(v_1v_2)^t = v_2v_1$ for $v_i \in V$. Then, the spinor norm is defined as $N(x) = x^t x$ for $x \in \mathrm{Cl}$.

Remark A.6. For a spanning element of $v = v_1 v_2 \dots v_n$, $N(v) = q(v_1)q(v_2)\dots q(v_n)$. Moreover, restriction of N to Γ_V is a homomorphism to $\mathbb{K}^\times$

[1] Here reflections refer to multiplying only one of the coordinates by -1, leaving the rest the same.

We start by considering Real Clifford algebras, and real pin, spin groups of them. We assume that the signature of quadratic form is either **Lorentzian** or **Euclidean**.

Definition A.7 (Real Pin Group)**.** Pin group is the kernel of $|N| : \Gamma_N \to \mathbb{R}^{>0}$.

Definition A.8 (Spin Group)**.** Spin group consists of the even elements of Pin, i.e. $\mathrm{Spin} = \mathrm{Cl}^0 \cap \mathrm{Pin}$

In particular, a Real Pin group fits into the following commutative diagrams where each nodes are exact.

$$\begin{array}{ccccccccc}
 & & 1 & & 1 & & 1 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \pm 1 & \longrightarrow & \mathrm{Pin} & \longrightarrow & O_V(\mathbb{R}) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \mathbb{R}^\times & \longrightarrow & \Gamma_V & \longrightarrow & O_V(\mathbb{R}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle |N|} & & \downarrow{\scriptstyle |N|} & & \downarrow & & \\
1 & \longrightarrow & \mathbb{R}^{>0} & \longrightarrow & \mathbb{R}^{>0} & \longrightarrow & 1 & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 1 & & 1 & & & &
\end{array} \tag{15}$$

As evident in this diagram, Pin is a **double cover** of $O_V(\mathbb{R})$. Similarly, it is not too hard to see that Spin is a double cover of the identity component of $O_V(\mathbb{R})$.

Definition A.9 ($\mathrm{Pin}_n^\pm$)**.** We define Pin_n^+ from a positive definite $\mathrm{Cl}_{0,n}$ and Pin_n^- from a negative definite $\mathrm{Cl}_{n,0}$. They define the group extensions

$$1 \to \{\pm 1\} \to \mathrm{Pin}_n^\pm \to O_n \to 1$$

Remark A.10. $\mathrm{Pin}^\pm$ can be distinguished from the fact that the lift of a reflection O_n squares to ± 1 respectively. In particular, $\mathrm{Pin}_1^+ = \mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathrm{Pin}_1^- = \mathbb{Z}_4$.

B Affine to Curved Geometry

B.1 Motivation: from Klein geometry to Cartan geometry

Before giving the formal definitions, it is worth isolating the guiding principle. Felix Klein's Erlangen program proposes that a geometry is specified by a homogeneous space G/H, where "points" are cosets, and geometric invariants which are the properties preserved by the transitive action of G. Affine space $\mathbb{A}^n$, with $G = \mathrm{Aff}_n$ and $H = \mathrm{GL}_n(\mathbb{R})$ is the basic example relevant here.

Every point of $\mathbb{A}^n$ looks exactly like every other point, related by a global symmetry of $\mathbb{A}^n$. The passage to curved geometry replaces this global symmetry by an infinitesimal one. A general manifold X need not admit any transitive symmetry group at all, but it is still be required to "look" like $\mathbb{A}^n$ at each point, with the identification twisting from point to point in a manner controlled by a connection. This is the sense in which a differential H_n-structure is the curved analogue of the Cartan symmetry type (H_n, ρ_n). The model geometry $\mathbb{A}^n$ is replaced by

X, the global group $\mathcal{G}_n \subset \mathrm{Aff}_n$ is replaced by a principal H_n-bundle, and the simply transitive translation action is replaced by the frame bundle together with a torsion-free connection. The rest of this appendix makes this precise.

B.2 Affine geometry and Cartan symmetry types

We begin with affine geometry. Let $\mathbb{A}^n$ denote an n-dimensional affine space. Unlike a vector space, an affine space does not have a distinguished origin, but it has a simply transitive action of its translation group $\mathbb{R}^n$. Its symmetry group is therefore the affine group

$$1 \longrightarrow \mathbb{R}^n \longrightarrow \mathrm{Aff}_n \longrightarrow \mathrm{GL}_n(\mathbb{R}) \longrightarrow 1.$$

More generally, a translationally invariant geometric structure on $\mathbb{A}^n$ has a symmetry Lie group $\mathcal{G}_n$ sitting in a map of extensions:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{R}^n & \longrightarrow & \mathcal{G}_n & \longrightarrow & H_n & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow{\scriptstyle \rho_n} & & \\ 1 & \longrightarrow & \mathbb{R}^n & \longrightarrow & \mathrm{Aff}_n & \longrightarrow & \mathrm{GL}_n(\mathbb{R}) & \longrightarrow & 1. \end{array}$$

With the pair

$$(H_n, \rho_n), \qquad \rho_n : H_n \longrightarrow \mathrm{GL}_n(\mathbb{R}),$$

defining the Cartan symmetry type of the geometric structure. Note $\mathcal{G}_n$ is the subgroup of Aff_n preserving the extra structure (a metric, a symplectic form, a complex structure, etc.), and H_n is its image in the linear group $\mathrm{GL}_n(\mathbb{R})$, i.e. the group of linear automorphisms of a single tangent space $T_p\mathbb{A}^n \cong \mathbb{R}^n$. In contrast to the quantum field theory setting, H_n need not be compact, and the target of ρ_n is $\mathrm{GL}_n(\mathbb{R})$ rather than O_n. We are no longer requiring a Wick-rotated reflection-positive structure, only a torsion-free geometric one. The basic examples are summarized by the following table:

H_n	Affine geometry	Curved geometry
O_n	Euclidean	Riemannian
$O_{1,n-1}$	Minkowskian	Lorentzian
$Sp_{n/2}(\mathbb{R})$	symplectic	symplectic
$GL_{n/2}(\mathbb{C})$	complex	complex
$U_{n/2}$	Hermitian	Kähler
Spin_n	spin	spin

In each row, ρ_n is the evident representation of H_n. Internal symmetry groups can also be incorporated into this framework, by taking a further product with a compact group acting trivially on $\mathbb{R}^n$. This is not carried out here or in [F].

B.3 Infinitesimal parallelism: frame bundles and connections

As stated in [F], the transition from affine to curved geometry can be understood in terms of parallelism. Affine geometry has a global parallelism, encoded by the simply transitive action of the translation group $\mathbb{R}^n$. That is, any two tangent spaces of $\mathbb{A}^n$ are canonically identified, since translation by the vector joining their basepoints carries one to the other. On a general manifold, there is no canonical way to identify tangent spaces at distinct points. The curved analogue of global parallelism is therefore "infinitesimal parallelism," encoded by an affine connection.

Definition B.1. Let X be a smooth n-manifold. The frame bundle $\mathcal{B}(X)$ is the principal $\mathrm{GL}_n(\mathbb{R})$-bundle whose fiber over $p \in X$ is

$$\mathcal{B}_p(X) = \{(e_1, \dots, e_n) \mid (e_1, \dots, e_n) \text{ is a basis of } T_pX\}.$$

The frame bundle carries a tautological $\mathbb{R}^n$-valued 1-form, the "soldering form":

$$\theta \in \Omega^1\big(\mathcal{B}(X); \mathbb{R}^n\big), \qquad \theta_{(p,e)}(v) = e^{-1}\big(d\pi(v)\big),$$

where $\pi : \mathcal{B}(X) \to X$ is the bundle projection and $e : \mathbb{R}^n \xrightarrow{\sim} T_pX$ is the frame regarded as a linear isomorphism. The soldering form is the mechanism by which $\mathcal{B}(X)$ "remembers" the tangent space at each point in a way that can be compared with the model space $\mathbb{R}^n$. That is, it plays the role that the tautological identification of tangent spaces plays in $\mathbb{A}^n$.

An affine connection on X is a connection 1-form on the principal $\mathrm{GL}_n(\mathbb{R})$-bundle $\mathcal{B}(X) \to X$, i.e. a $\mathfrak{gl}_n(\mathbb{R})$-valued 1-form on $\mathcal{B}(X)$ satisfying the usual equivariance and reproducing properties. Given such a connection ω, its torsion is the $\mathbb{R}^n$-valued 2-form

$$T = d\theta + \omega \wedge \theta,$$

and the connection is torsion-free if $T = 0$. Torsion-free connections are exactly those whose parallel transport, to second order, agrees with translation in the model affine space $\mathbb{A}^n$, matching the fact that Aff_n itself has no "twisting" built into its translation subgroup. In the special case where the image of ρ_n lies in O_n (i.e. where the structure includes a Riemannian metric) the resulting geometry is Riemannian, and the unique torsionfree compatible connection is the Levi-Civita connection. This is the classical fundamental theorem of Riemannian geometry, and it is the model for the general existence and uniqueness statements one looks for in H_n-geometry.

B.4 Differential H_n-structures

We can now replace the global affine symmetry by a local geometric structure on a curved manifold. Let (H_n, ρ_n) be a Cartan symmetry type, where

$$\rho_n : H_n \longrightarrow \mathrm{GL}_n(\mathbb{R})$$

is a homomorphism. The group H_n specifies the allowed local symmetries, while ρ_n describes how these symmetries act on the tangent space.

Definition B.2 (Differential H_n-structure)**.** Let (H_n, ρ_n) be a Cartan symmetry type. A differential H_n-structure on a smooth n-manifold X is a triple

$$(P, \Theta, \varphi),$$

where $P \to X$ is a principal H_n-bundle equipped with a connection Θ, and

$$\varphi : \mathcal{B}(X) \longrightarrow \rho_n(P)$$

is an isomorphism of principal $\mathrm{GL}_n(\mathbb{R})$-bundles such that the induced connection

$$\varphi^* \rho_n(\Theta)$$

is torsion-free.

Here $\rho_n(P) = P \times_{H_n} \mathrm{GL}_n(\mathbb{R})$ is the principal $\mathrm{GL}_n(\mathbb{R})$-bundle induced from P along ρ_n, and $\rho_n(\Theta)$ is the induced connection on it. Two remarks clarify the roles of the data (P, Θ, φ):

Remark B.3.

- The isomorphism φ is a reduction of structure group. Since $\mathcal{B}(X)$ is the frame bundle of X, an isomorphism $\varphi : \mathcal{B}(X) \to \rho_n(P)$ of $\mathrm{GL}_n(\mathbb{R})$-bundles is the classical data of a reduction of the structure group of X from $\mathrm{GL}_n(\mathbb{R})$ to H_n (via ρ_n), together with a choice of identification. When ρ_n is injective this recovers the familiar statement that, e.g., a Riemannian metric is equivalently an O_n-reduction of the frame bundle. Note φ picks out, at each point, the subset of frames that are "orthonormal." When ρ_n is not injective, the same data still makes sense and encodes the additional lift information e.g. a spin structure.

- The torsion-free condition links the connection to the smooth structure of X itself. The connection Θ on P a priori has nothing to do with the tangent bundle of X. It becomes geometric only once transported to $\mathcal{B}(X)$ via φ and is required to be torsion-free. This is the curved analogue of the requirement in the affine picture that $\mathcal{H}_n$ sits over H_n compatibly with the fixed inclusion $H_n \hookrightarrow \mathrm{GL}_n(\mathbb{R}) \hookrightarrow \mathrm{Aff}_n$. The "rotational" part H_n and the "translational" part $\mathbb{R}^n$ are locked together, and $\varphi^*\rho_n(\Theta)$ torsion-free is exactly what locks them together on X.

Example B.4 (Riemannian structures, worked out)**.** Take $H_n = O_n$ with ρ_n the standard representation. A differential H_n-structure on X then consists of:

- a principal O_n-bundle $P \to X$ with connection Θ;
- an isomorphism $\varphi : \mathcal{B}(X) \to \rho_n(P)$, torsion-free for Θ.

The associated vector bundle $P \times_{O_n} \mathbb{R}^n$ carries a canonical fiber-wise inner product (since O_n preserves the standard inner product on $\mathbb{R}^n$), and φ transports this inner product to TX producing a Riemannian metric g on X. Conversely, given g, let P be the bundle of g-orthonormal frames; the torsion-free, metric-compatible connection Θ on P is then forced to be the Levi-Civita connection by the fundamental theorem of Riemannian geometry, and φ is the tautological inclusion $\mathcal{B}(X) \supset P \hookrightarrow \rho_n(P)$ composed with the isomorphism $\rho_n(P) \cong \mathcal{B}(X)$ induced by $O_n \hookrightarrow \mathrm{GL}_n(\mathbb{R})$. Thus for $H_n = O_n$ the abstract definition recovers, and is equivalent to, an ordinary Riemannian metric together with its Levi-Civita connection.

Remark B.5. Existence of a differential H_n-structure on X is a topological question requiring the classifying map $X \to B\,\mathrm{GL}_n(\mathbb{R})$ to lift along $B\rho_n : BH_n \to B\,\mathrm{GL}_n(\mathbb{R})$. The torsion-free condition is typically a matter of existence and uniqueness of a compatible connection, exactly as in the Riemannian case above. We will not need the general existence theory here since it is enough to know that a differential H_n-structure is the natural curved refinement of a Cartan symmetry type (H_n, ρ_n).

Thus, the affine picture provides the model geometry and its global symmetry group, while a differential H_n-structure replaces this global symmetry by an infinitesimal symmetry acting on each tangent space, mediated by the frame bundle and its soldering form. The study of H_n-geometry is then the study of the invariants of smooth manifolds equipped with such structures and their diffeomorphisms. It is this notion that is used in the main text to build the bordism category $\mathrm{Bord}_{\langle n-1,n\rangle}(H_n^{\nabla})$ underlying Axiom 1 where objects and morphisms of that category are manifolds equipped with differential H_n-structures as defined here.

6 References

Sources.

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